

PHASE 4 – Incident Response & Automation

Incident Response Implementation for HFT SOC Environment

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1. Incident Scenario Description

Selected Incident Scenario

Unauthorized Trader Account Compromise Leading to Trading Algorithm Manipulation

Organization Context

The organization is an Automated High Frequency Trading (HFT) firm where trading decisions are executed by automated algorithms running on low latency trading servers. The SOC continuously monitors infrastructure using SIEM tools to detect abnormal behavior that may impact trading integrity.

Since the organization depends heavily on algorithm integrity and trader account security, any unauthorized access can lead to financial loss, market instability, and regulatory violations.

The SOC uses:

SIEM platform → Wazuh

Log sources → Linux servers, Windows workstations, firewall, AD logs

Detection → correlation rules from Phase-2

Threat intelligence → Phase-3 CTI sources

How the Incident Started

The incident started with a phishing email sent to a trader pretending to be a market report notification. The trader clicked the link and entered credentials into a fake login portal.

The attacker obtained trader credentials and attempted multiple logins from an external IP address.

After successful login attacker:

- Accessed trader workstation
- Attempted algorithm parameter modification
- Executed suspicious scripts
- Attempted log deletion

This represents a realistic SOC incident because credential compromise is one of the most common enterprise attacks.

Systems Affected

Affected systems include:

Trader workstation (Windows)
Active Directory authentication server
Trading algorithm server (Linux)
SIEM monitoring platform
Firewall logs

Indirectly affected:

Order execution engine
Risk management server

Because algorithm manipulation could affect trade execution.

Initial Symptoms (SOC Alerts & Logs)

SOC detected anomalies through:

SIEM Alerts:

Multiple failed logins detected
Login from unusual IP
Privileged access after failed login
File modification alert
Suspicious process execution

Log Evidence:

Windows Event ID 4625 failed login
Linux file integrity alerts
EDR suspicious process alerts
Firewall IP anomaly logs

Initial SOC Alert:

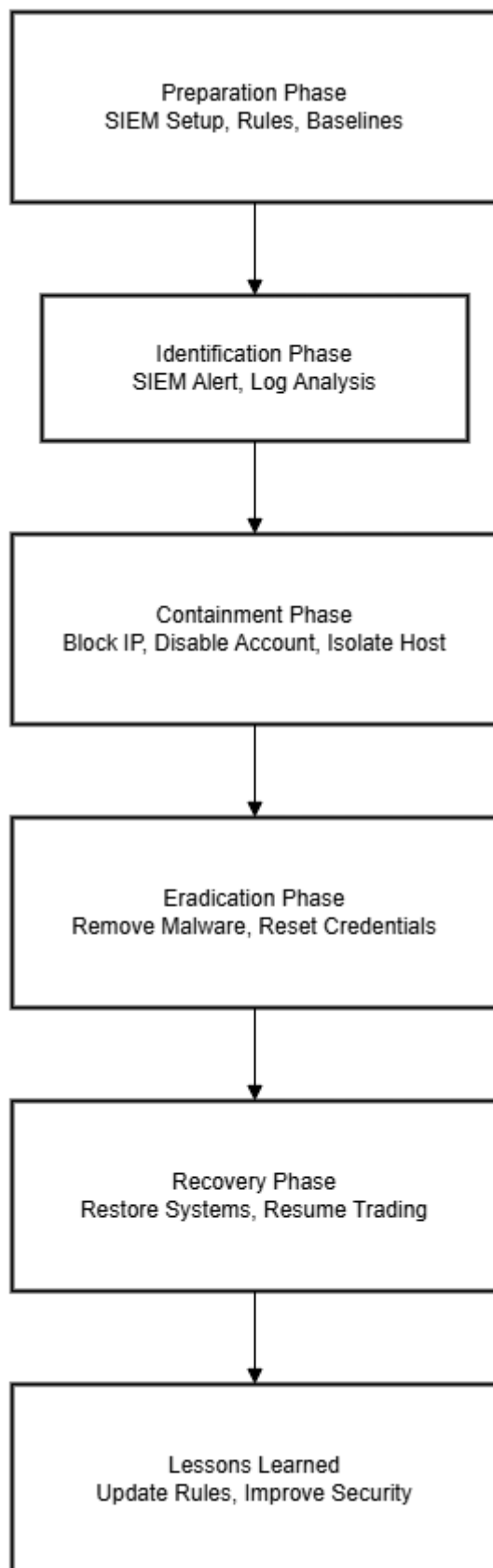
HIGH severity alert triggered due to correlation rule:

Failed logins

- New IP login
- Privileged activity

This triggered Tier-1 investigation.

2. Incident Response Lifecycle



IR Lifecycle Explanation

Phase 1 – Preparation

Activities performed:

SIEM configured with detection rules
Log collection enabled
EDR monitoring enabled
Incident response playbook prepared
SOC escalation matrix defined

Responsible:

SOC Engineer
Tier-3 SOC

Evidence Prepared:

Baseline login behavior
Trading algorithm file hashes
Authorized user list

Goal:

Ensure SOC readiness before incident.

Phase 2 – Identification

SOC detects incident through:

SIEM alert correlation
Authentication anomaly detection
File integrity alerts

Tier-1 Actions:

Validate alert
Check IP reputation
Verify user activity
Confirm anomaly

Evidence collected:

Authentication logs
Firewall logs
SIEM alerts
User activity logs

Responsible:

Tier-1 SOC analyst

Goal:

Confirm whether incident is true positive.

Phase 3 – Containment

After confirming compromise:

Actions taken:

Disable compromised trader account

Block malicious IP

Isolate workstation

Stop suspicious processes

Prevent algorithm modification

Evidence collected:

Memory artifacts

Process logs

Network connections

File modification logs

Responsible:

Tier-2 SOC

IR Lead

Goal:

Prevent further damage.

Phase 4 – Eradication

Root cause removal:

Remove malware

Remove persistence scripts

Reset credentials

Patch system

Remove unauthorized accounts

Evidence:

Malicious hashes

Script artifacts

Persistence registry entries

Responsible:

Tier-3 SOC

SOC Engineer

Goal:

Remove attacker presence completely.

Phase 5 – Recovery

System restoration:

- Restore clean system state
- Re-enable trader account
- Verify algorithm integrity
- Resume trading operations
- Monitor for recurrence

Responsible:

- Tier-3 SOC
- Infrastructure team

Evidence:

- System integrity reports
- Post recovery monitoring logs

Goal:

Return to normal operations safely.

Phase 6 – Lessons Learned

Post incident improvements:

- Improve phishing awareness
- Improve SIEM detection rule
- Add MFA for traders
- Improve alert thresholds
- Improve SOC response time

Responsible:

- SOC Manager
- SOC Engineers

Goal:

Prevent recurrence.

3. Severity & Prioritization Matrix

SOC Severity Classification Matrix

Severity	Impact	Urgency	Example	Response Time
Low	Minimal impact	Low	Single failed login	24 hours
Medium	Limited system impact	Medium	Malware alert	4 hours
High	Critical asset affected	High	Account compromise	1 hour
Critical	Business disruption	Immediate	Algorithm manipulation	Immediate

Incident Severity Assignment

Selected incident severity:

HIGH → Because:

Trader account compromised
Algorithm modification attempt
Critical asset affected

If algorithm changed:

Severity becomes CRITICAL.

Justification

This incident is classified HIGH because:

Access to trading account obtained
Integrity risk exists
Financial impact possible
Regulatory impact possible

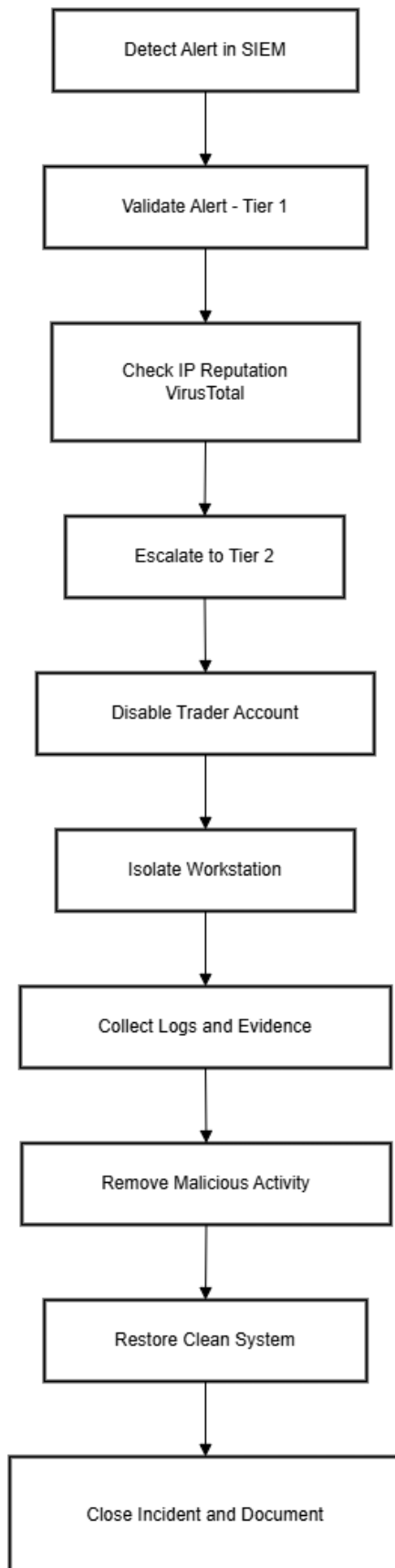
Urgency is high because:

Trading environments are time sensitive.

Response time required:

Within 1 hour.

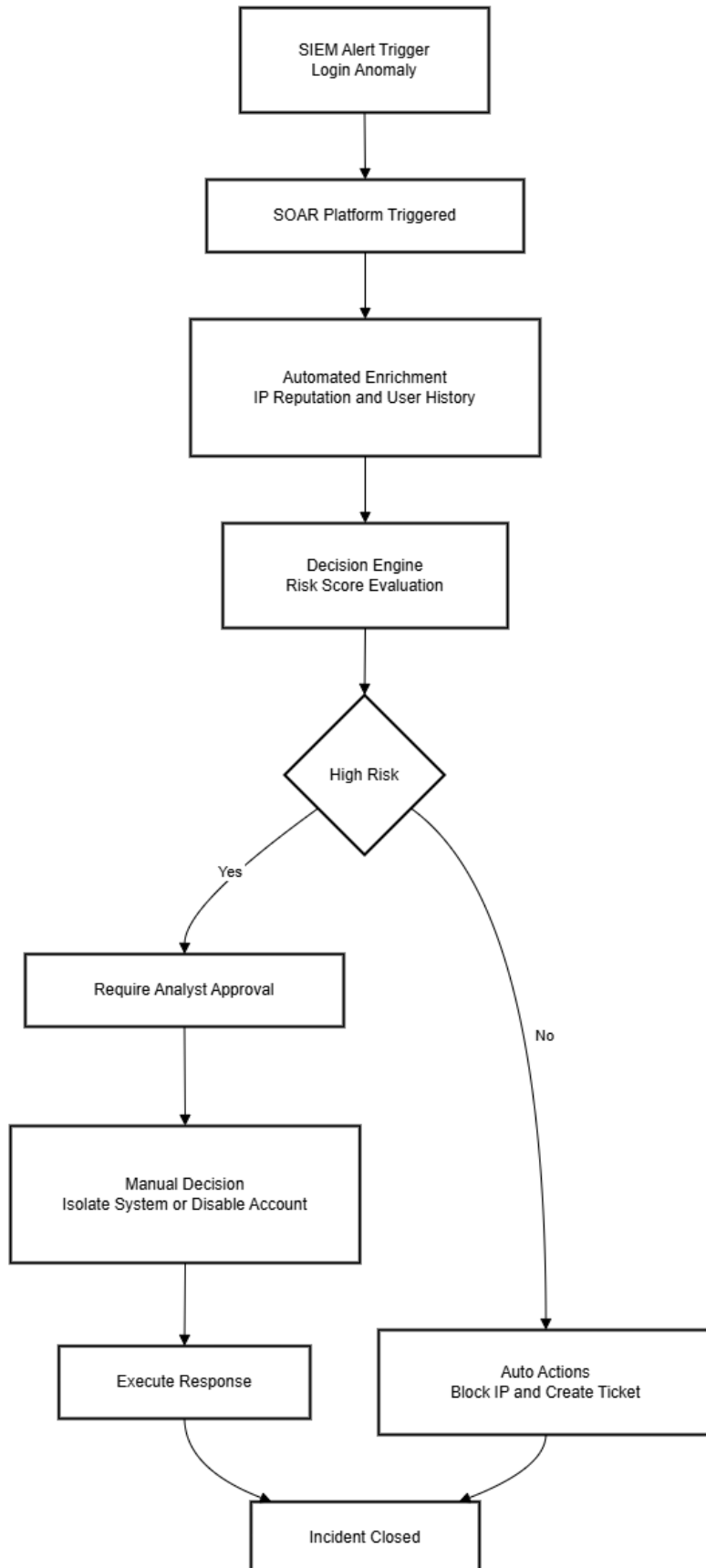
4. Incident Response Playbook



IR Playbook

Step	Action	Responsible	Tools	Expected Outcome
1	SIEM alert detected	Tier-1	Wazuh	Alert identified
2	Validate login anomaly	Tier-1	SIEM logs	Confirm suspicious activity
3	Check IP reputation	Tier-1	VirusTotal	Malicious IP confirmed
4	Escalate incident	Tier-1	SOC workflow	Tier-2 engaged
5	Disable account	Tier-2	AD tools	Access blocked
6	Isolate workstation	Tier-2	EDR	Spread prevented
7	Collect forensic logs	Tier-2	Syslogs	Evidence preserved
8	Remove malicious files	Tier-3	EDR	Threat removed
9	Restore clean state	Tier-3	Backup	System recovered
10	Close incident	Tier-3	SIEM	Incident resolved

5. SOAR Automation Flow



SOAR Workflow Explanation

Alert Trigger

SIEM detects:

Multiple failed logins

Suspicious login IP

Automatically triggers SOAR workflow.

Automated Enrichment

SOAR automatically collects:

IP reputation

User history

Login geolocation

Process history

Threat intelligence data

Tools:

VirusTotal

Threat feeds

SIEM context data

Outcome:

Context enrichment without analyst effort.

Decision Logic

SOAR evaluates:

Login anomaly score

User behavior deviation

IP risk score

If risk high:

Escalate to analyst.

Automated Response

Possible automated actions:

Block IP

Disable account

Create SOC ticket
Notify SOC team
Start investigation workflow

Analyst Approval Required

Human approval needed for:

Account disabling
Server isolation
Trading halt decision
Algorithm rollback

Reason:

Avoid business disruption.

Benefits of SOAR

Advantages:

Faster detection
Consistent response
Reduced SOC workload
Automated enrichment
Reduced MTTD
Reduced MTTR

Most useful automation steps:

Alert enrichment
IP blocking
Ticket creation
Evidence collection

Human required:

Business decisions
Containment approval
Recovery approval

Final Conclusion

This phase demonstrated practical SOC incident response using a realistic HFT trading security incident. The project shows how SOC teams detect, investigate, contain, and recover from incidents using structured IR lifecycle processes.

The integration of SOAR concepts demonstrates how automation can improve response speed, reduce manual workload, and improve SOC efficiency while maintaining human oversight for critical business decisions.

This phase completes the SOC operational lifecycle from detection to response and automation.